KP01 XOR and the limit of one line

> Level: undergraduate (year 3)
>
> Pattern Recognition | Chapter 4, Sections 4.1 to 4.2

Slides (view and download)

Learning objectives

- Turn the truth tables of OR, AND and XOR into classification problems in the plane.
- Compute each prediction from a score and a fixed rule, and check it against the label.
- Use the sums of two A scores and of two B scores to prove that one line cannot separate XOR, and tell a representational limit from a training problem.

KP01 XOR and the limit of one line

One-sentence summary: For the same four points one line separates OR and AND but not XOR: the two A scores must sum to a positive number and the two B scores to at most 0, yet it is the same expression; more training cannot fix it, so what must change is the combination of boundaries.

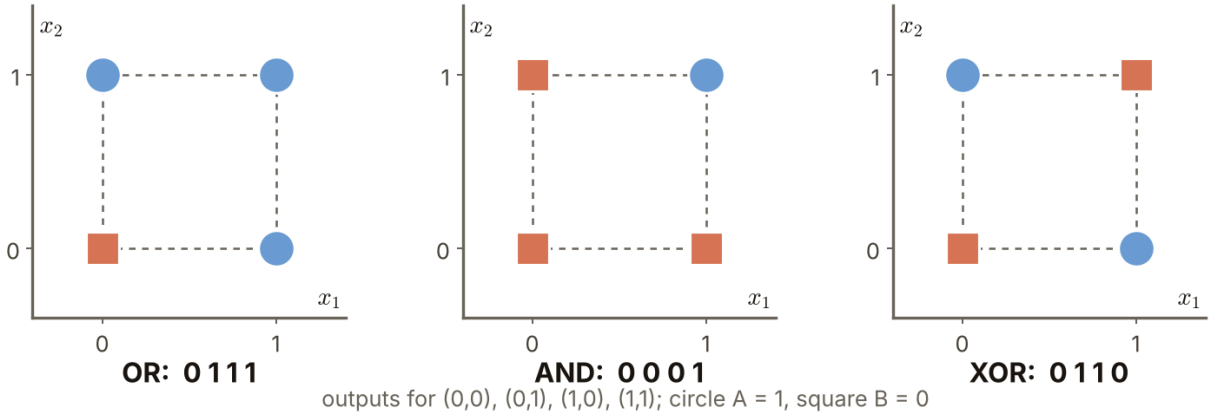
1. Same four points, new labels: is one line enough?

Plain explanation: Chapter 3 let the model decide the class with one line. Here we keep the model and only change the labels of four points, to see whether it can still answer.

- **Label**: the class a point must be assigned to: A (circle, output 1) or B (square, output 0).
- **XOR (exclusive or)**: the output is 1 (A) only when the two inputs differ, and 0 (B) when they match. OR outputs 1 when at least one input is 1, and AND when both

inputs are 1.

Figure `lec05-kp01-three-labels`: The four inputs are $(0,0), (0,1), (1,0), (1,1)$, each input being 0 or 1. The three panels use the same coordinates with different labels; under each is the output in that order: OR 0111, AND 0001, XOR 0110.



For OR the class A occupies three corners and for AND only the upper-right corner, so one line seems enough; XOR puts each class on a diagonal, and we are not sure.

2. Give every point a score: positive means class A

Formal definition: We consider binary inputs and two classes. The score with a constant term and the prediction rule are (strictly this is an affine function, but by convention the classifier is still called linear):

$$g(\mathbf{x}) = w_1 x_1 + w_2 x_2 + w_0, \qquad \hat{y}(\mathbf{x}) = \begin{cases} 1\ (A), & g(\mathbf{x}) > 0, \\ 0\ (B), & g(\mathbf{x}) \le 0. \end{cases}$$

w_1, w_2 are coefficients and w_0 is a constant that shifts the line. A positive score gives class A and anything else gives B, including exactly 0; the tie rule must not depend on the true label. The set $g = 0$ is the line in the figure.

Linearly separable: some coefficients classify every point correctly by this rule. What matters is that such coefficients exist, not only how the current ones perform.

Example: For OR take $w_1 = w_2 = 1,\ w_0 = -0.5$, that is $g = x_1 + x_2 - 0.5$. The four scores are $-0.5, 0.5, 0.5, 1.5$; the last three are positive, so they are class A, exactly OR.

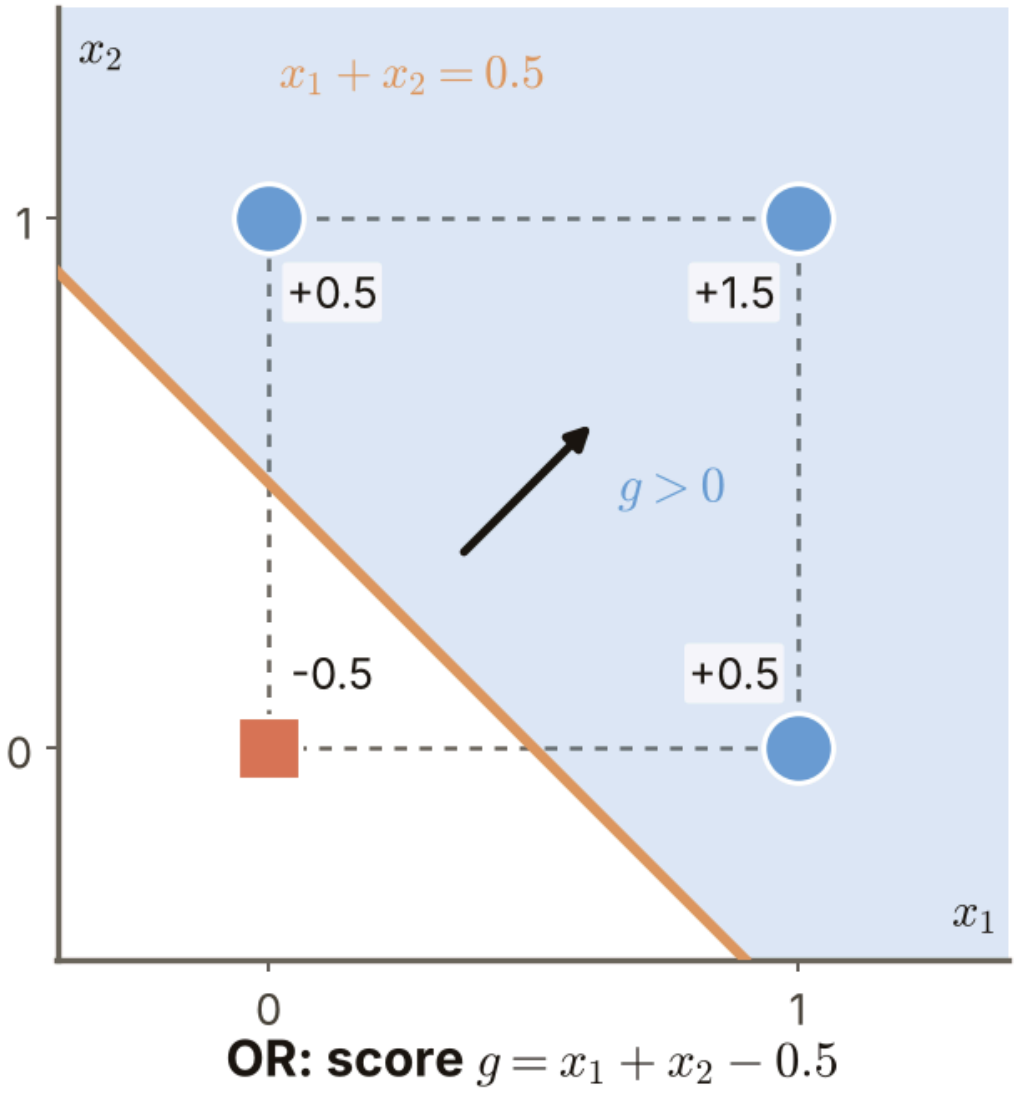


Figure `lec05-kp01-or-line`: Look at the boundary $x_1 + x_2 = 0.5$ first and then at the tinted positive side (arrow); each point shows its score, and all three A points are on the positive side.

3. The same line: OR all right, XOR one wrong

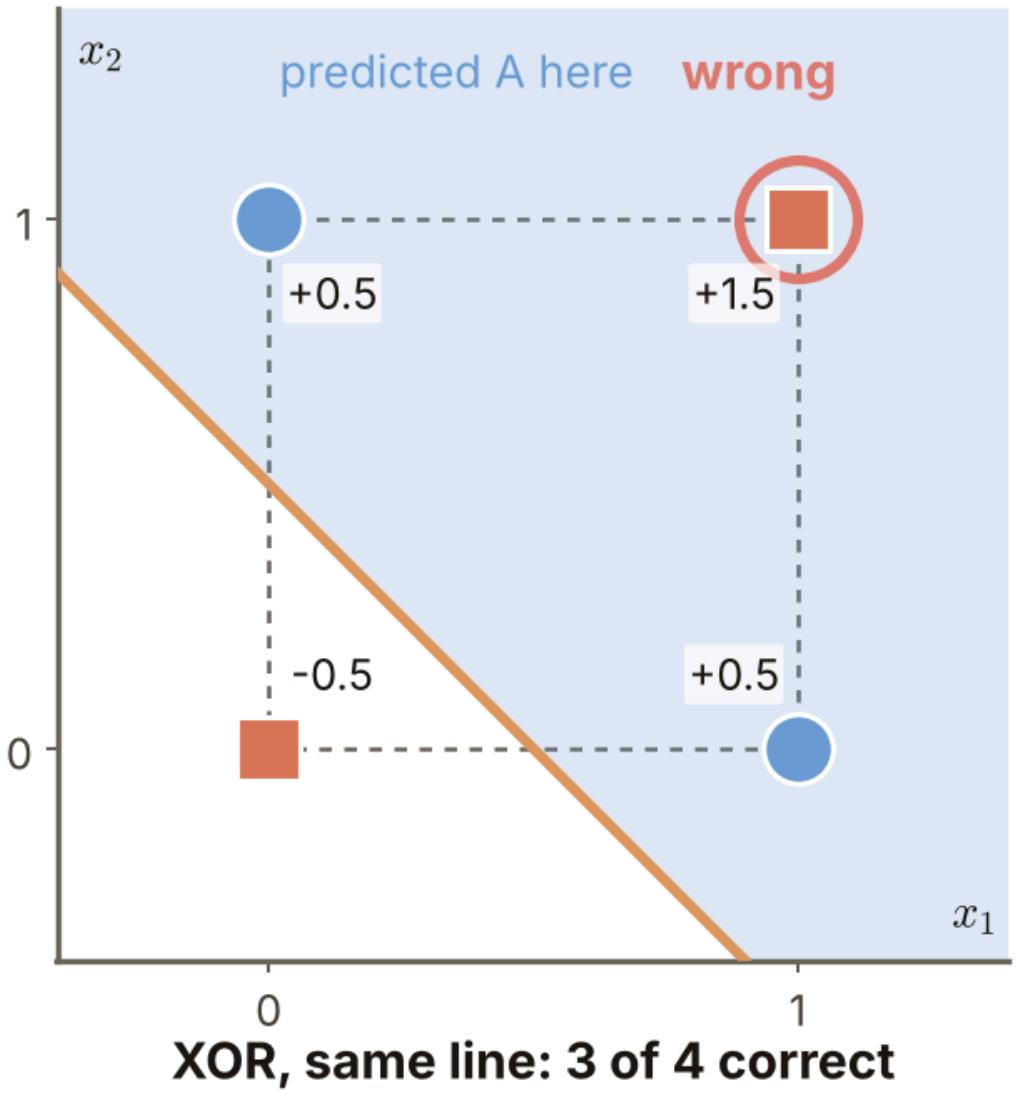
Use the same line on XOR and substitute each point into the same $g = x_1 + x_2 - 0.5$:

Input	Substitution	Score	Prediction	XOR truth
$(0, 0)$	$0 + 0 - 0.5$	-0.5	B	B
$(0, 1)$	$0 + 1 - 0.5$	0.5	A	A
$(1, 0)$	$1 + 0 - 0.5$	0.5	A	A
$(1, 1)$	$1 + 1 - 0.5$	1.5	A	B ×

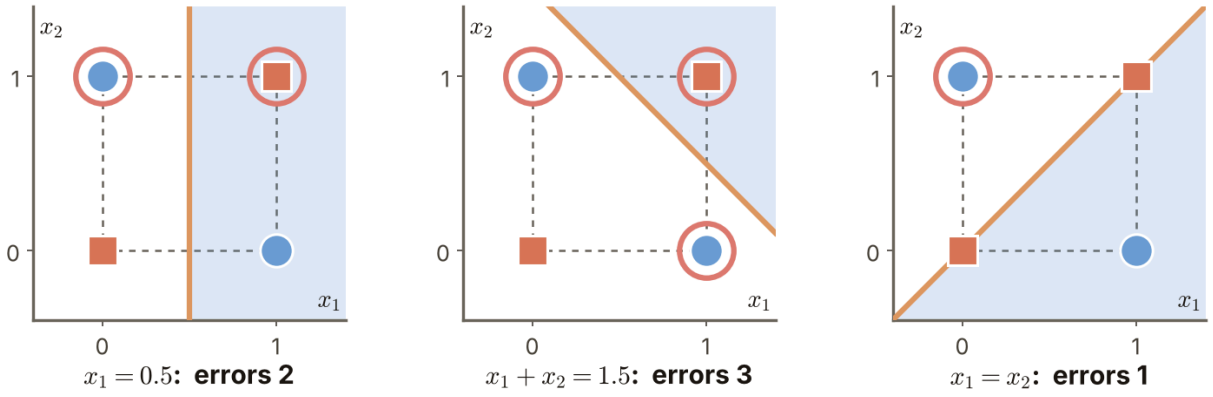
The predictions equal OR; for XOR one point is wrong:

$$\text{accuracy} = \frac{3}{4} = 0.75 = 75\%, \qquad 75\% + 25\% = 100\% \text{ (check)}.$$

This is the result of one line only and does not rule out other lines.



4. Three other lines: 2, 3 and 1 errors



blue tint = predicted A; red ring = wrong prediction

Score	Predictions (in the order $(0,0), (0,1), (1,0), (1,1)$)	Errors
$x_1 - 0.5$	$(0, 0, 1, 1)$	2
$x_1 + x_2 - 1.5$	$(0, 0, 0, 1)$	3
$x_1 - x_2$	$(0, 0, 1, 0)$	1

The second line cuts off only the upper-right corner, but that side is predicted A, so the errors are two A points and one B point. On the third line $x_1 = x_2$ two points have score 0 and are predicted B by the rule. Three failed lines are **not a proof**: there are too many lines we did not try.

5. The two A scores must add up to a positive number

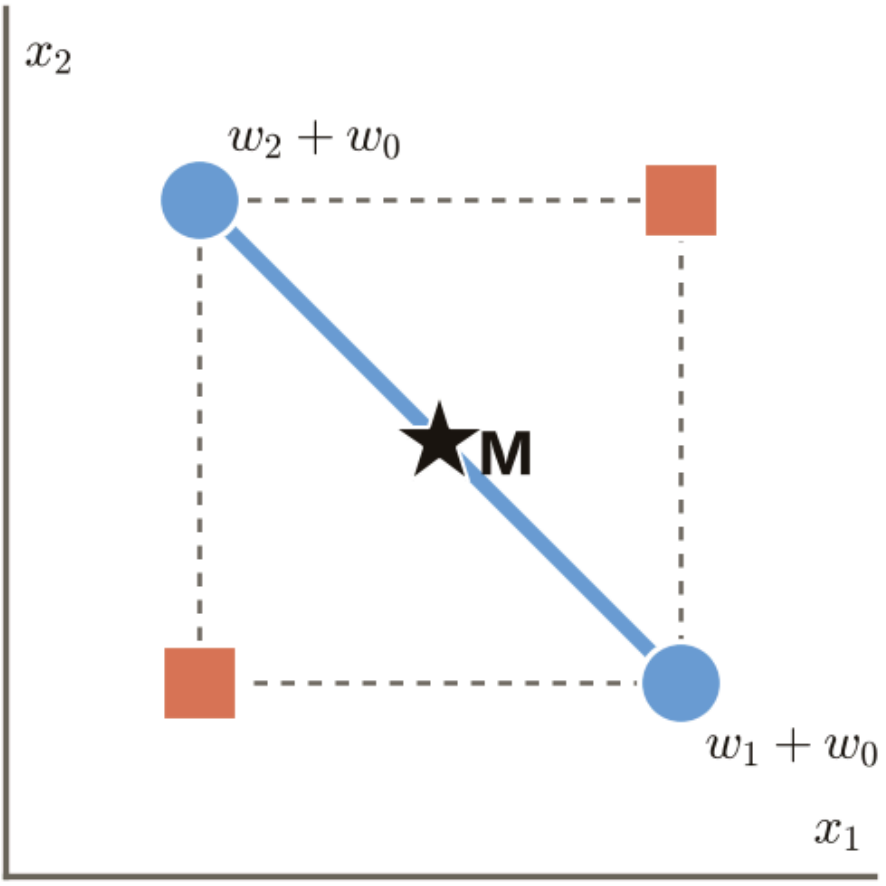
To rule out every line at once, assume some w_1, w_2, w_0 classify all of XOR correctly and write down what they must satisfy. For the two A points:

$$g(0,1) = w_2 + w_0 > 0, \qquad g(1,0) = w_1 + w_0 > 0.$$

Positive plus positive is positive, so adding the two gives:

$$(w_2 + w_0) + (w_1 + w_0) = w_1 + w_2 + 2w_0 > 0.$$

Geometrically the two A points form a diagonal with midpoint $M = (\frac{1}{2}, \frac{1}{2})$. Since g is linear, $g(M) = \frac{w_1 + w_2 + 2w_0}{2}$ is exactly the average of the two scores and is positive too.



$$g(M) = \frac{(w_2 + w_0) + (w_1 + w_0)}{2}$$

$M = (\frac{1}{2}, \frac{1}{2})$, midpoint of the two A points

6. The B scores add up to at most 0: a contradiction

For the two B points the scores must be at most 0 (by the fixed rule):

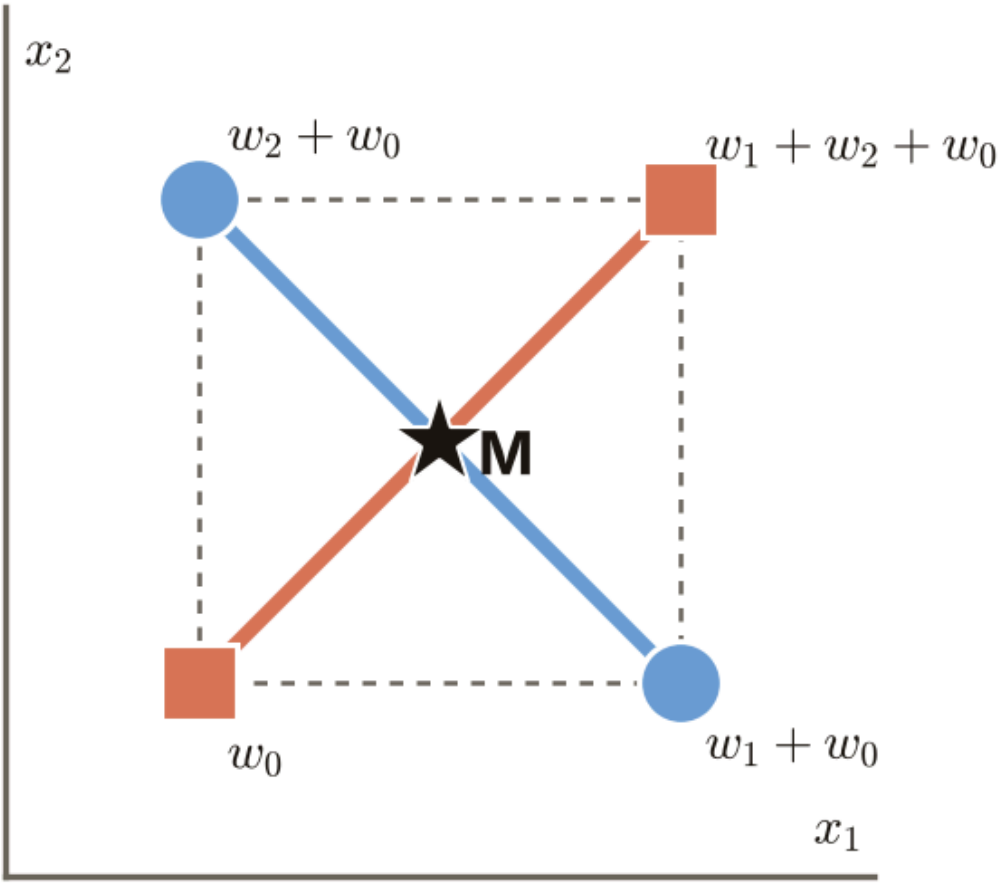
$$g(0,0) = w_0 \leq 0, \qquad g(1,1) = w_1 + w_2 + w_0 \leq 0.$$

A sum of at most 0 numbers cannot be positive:

$$w_0 + (w_1 + w_2 + w_0) = w_1 + w_2 + 2w_0 \leq 0.$$

Both sums are the **same expression**; A demands that it be greater than 0 and B that it be at most 0, a contradiction. Hence no coefficients classify every point correctly: this rules out every line, not just the plotted candidates. Geometrically the two diagonals

share the midpoint M: $g(M)$ would be the average of the A scores and also the average of the B scores, impossible to be both positive and at most 0.



$$g(M) = \frac{w_1 + w_2 + 2w_0}{2} \text{ for both diagonals}$$

average of A scores = average of B scores

7. More training lowers the error but cannot reach zero

Common misconception: More iterations, a smaller learning rate or a new initialization should eventually get everything right. These only search the same family of lines; they can turn 3 errors into 1, but that family has no perfect answer, so the lowest error is 1, not 0.

Representational capacity: whether the shape of the boundary can express the answer at all. Whether training finds the best line and whether the best line is good enough are different questions.

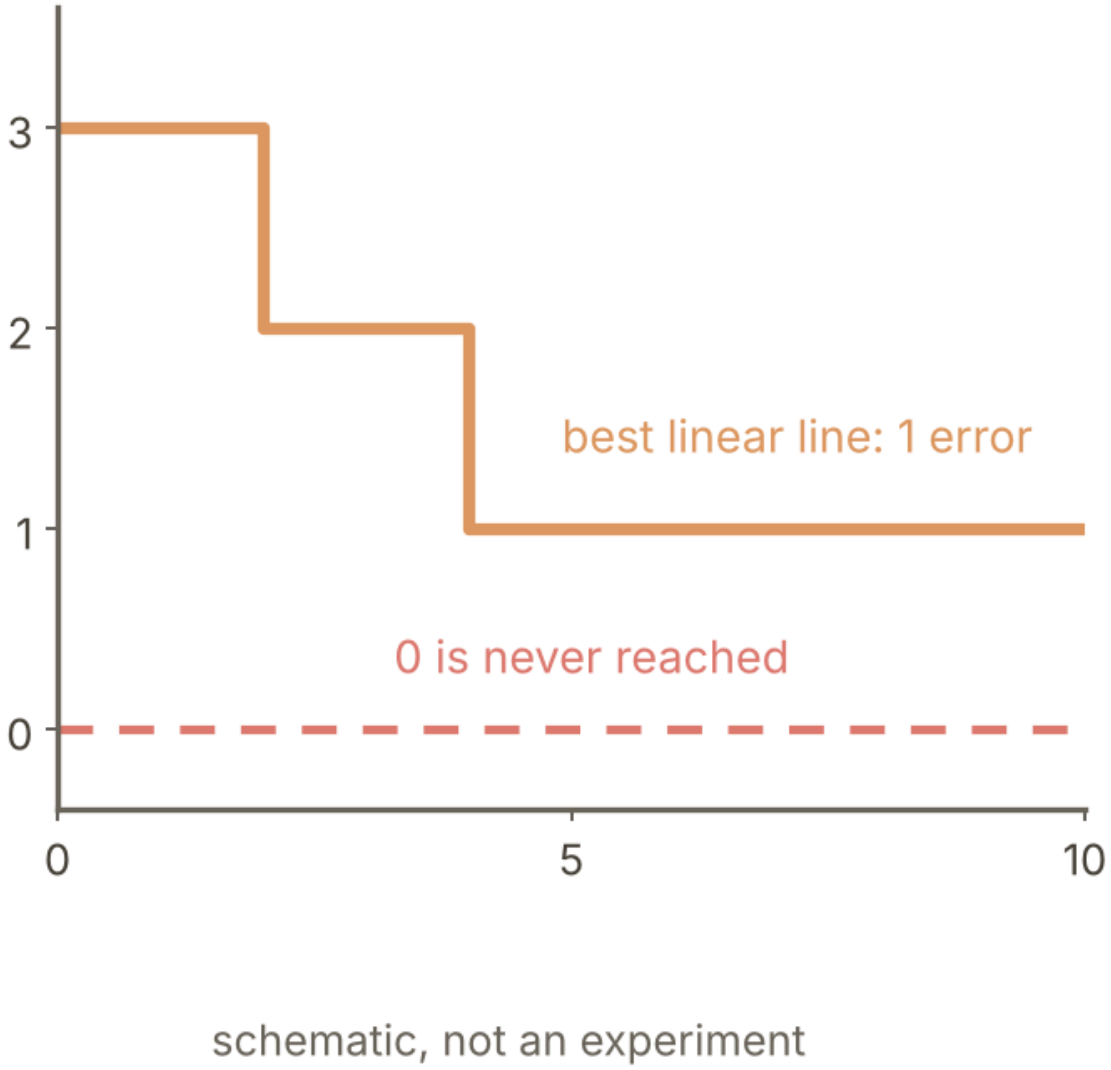


Figure `lec05-kp01-best`: This is a schematic, not an experiment; the dashed line is 0, which is never reached.

8. Change the boundaries: two lines make a strip

The proof rules out a single linear score with a fixed threshold, not combinations of boundaries. [Added] Let $s = x_1 + x_2$. The four XOR points have $s = 0, 1, 2$ and A is exactly $s = 1$, so the intersection of two decisions selects it:

$$\hat{y} = 1 \quad \Longleftrightarrow \quad 0.5 < s \leq 1.5.$$

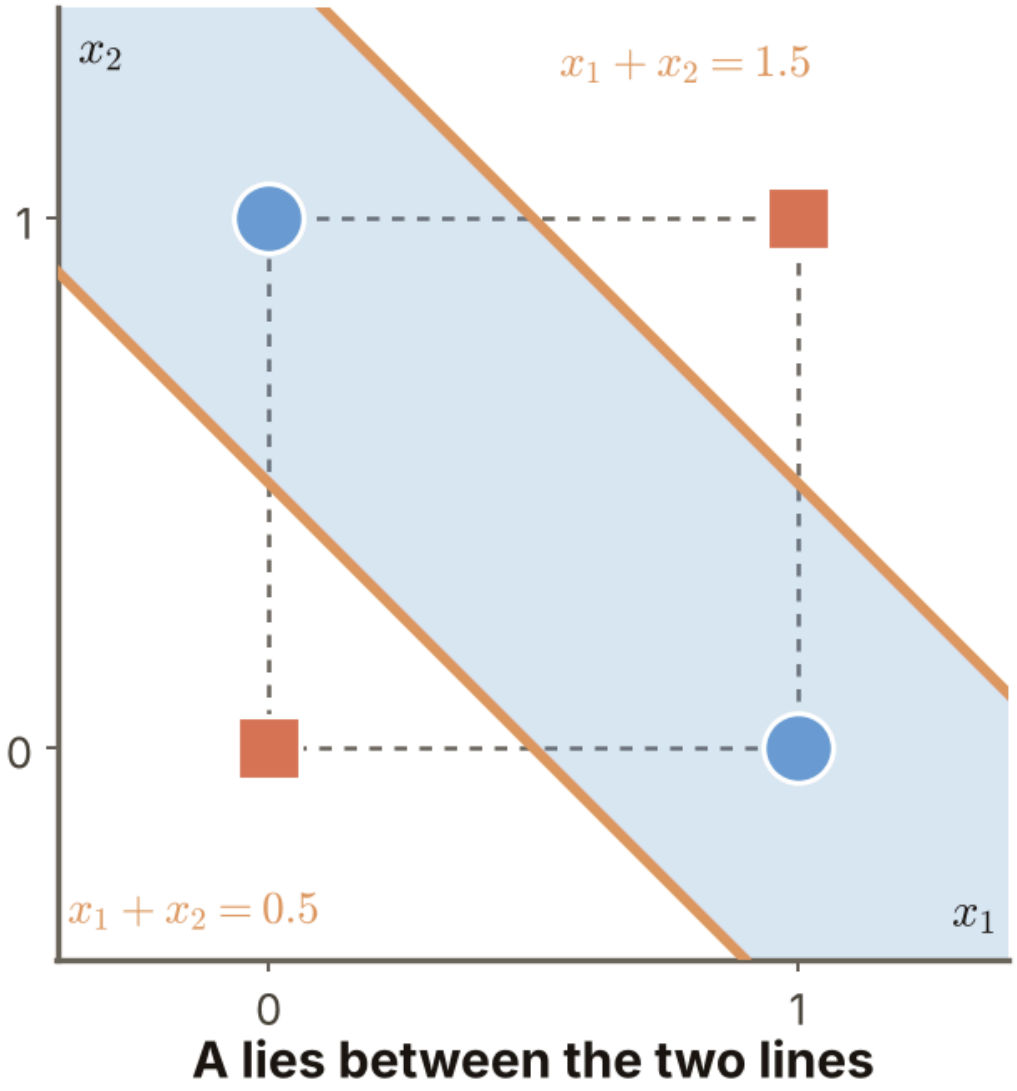
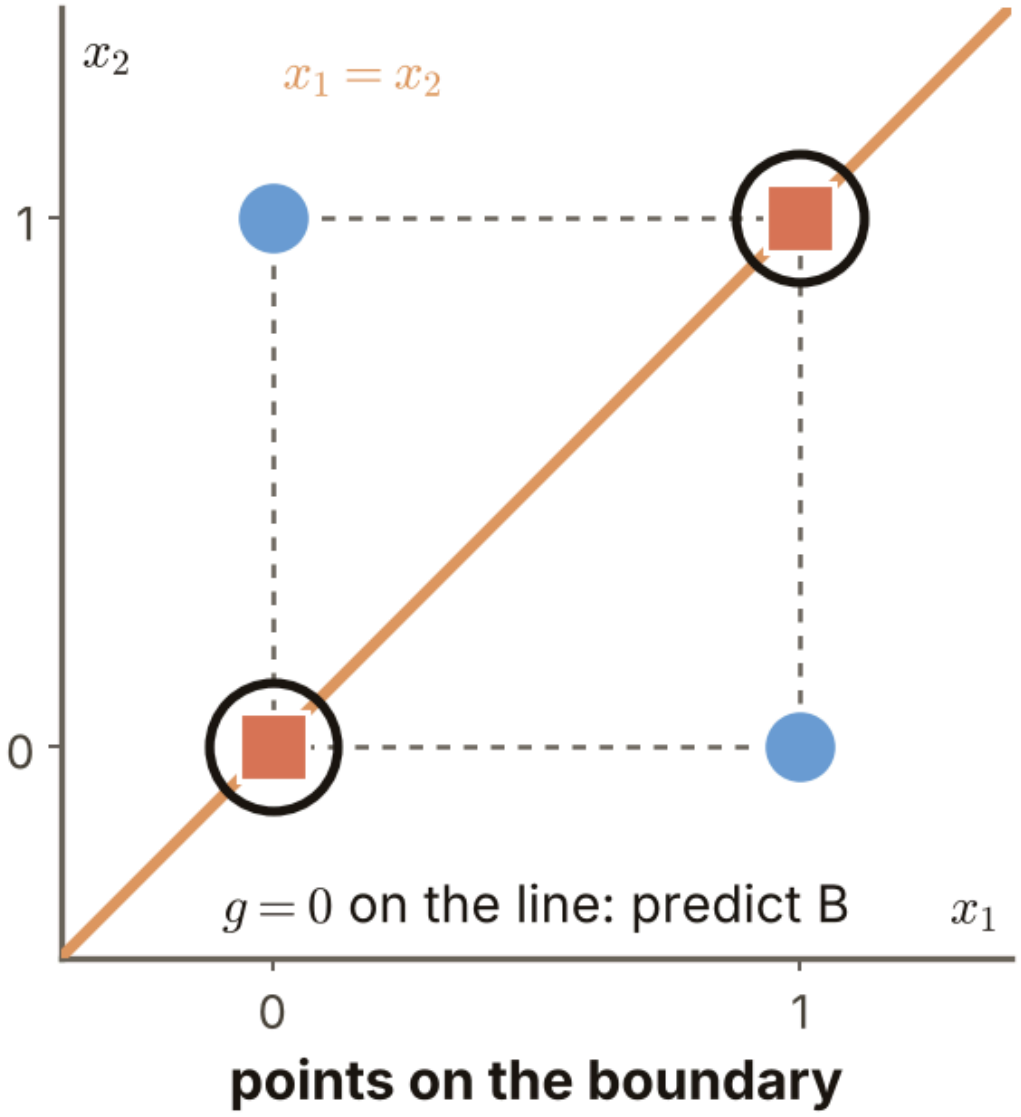


Figure `lec05-kp01-strip`: Two parallel lines enclose a slanted strip with both A points inside and both B points outside. This verifies only the four Boolean points and does not define a unique extension of XOR to the continuous plane.

What changes is the combination of boundaries a model may use, not another choice of old coefficients. The next KP asks how neurons can carry out these two decisions.

Extension: practice and implementation

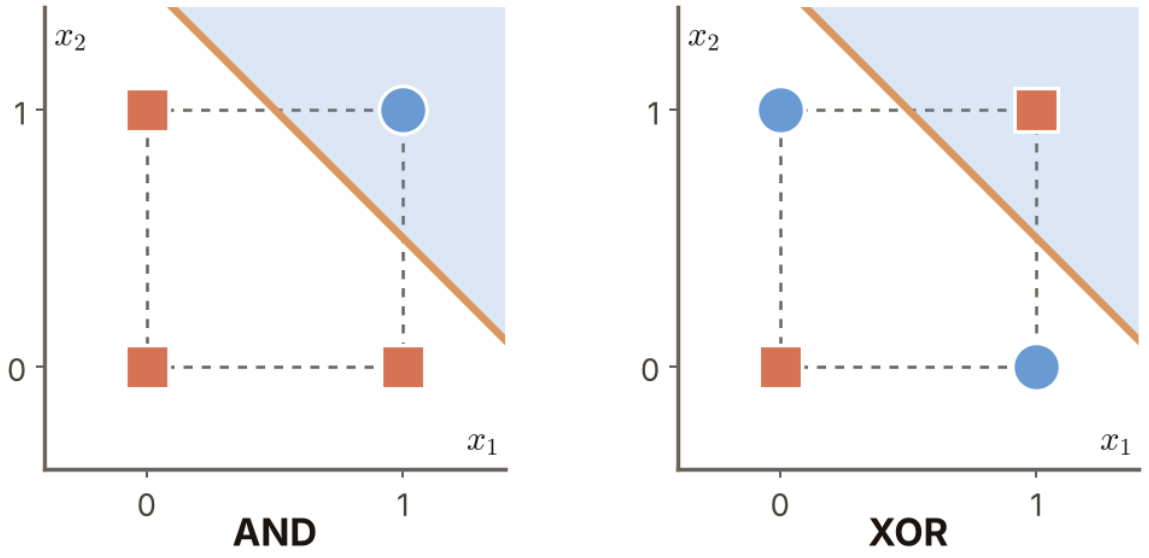
Code: Compute each score, predict with "score > 0" and compare with the true labels. Include boundary points in tests so that code and text use the same inequality. The half-integers here are exactly representable; general real data still need care with floating-point errors.



Practice:

1. Compute the four inputs with $g = x_1 + x_2 - 1.5$ and check AND and XOR separately. Hint: compute the scores, apply the same rule $g > 0$, and only then compare the two sets of labels.
2. Is XOR failing because of few samples, noise, learning rate, or the restricted boundary shape? Hint: which does the contradiction rule out?
3. Which of OR, XOR and XNOR is separable by one line? XNOR outputs 1 when the inputs match. Hint: draw the labels first.

Worked solution (question 1): ① Formula $g = x_1 + x_2 - 1.5$, class A if $g > 0$. ② Substitute the four points $(0, 0), (0, 1), (1, 0), (1, 1)$: the scores are $-1.5, -0.5, -0.5, 0.5$. ③ Only $(1, 1)$ is positive, so the predictions are B, B, B, A. ④ Check: AND is B, B, B, A, all correct; for XOR the points $(0, 1)$ and $(1, 0)$ are predicted B and $(1, 1)$ is predicted A, 3 errors.



line $x_1 + x_2 = 1.5$; blue tint = predicted A

Terminology and source

Term	Meaning
Label	the class a point must be assigned to
XOR	output 1 only when the two inputs differ
Score, constant	the number g obtained by substitution; the term that shifts the line
Linearly separable	some coefficients classify every point correctly
Representational capacity	whether the boundary shape can express the answer

Source: Chapter 4, KP01, Sections 4.1 to 4.2, accessed 2026-10-24. The four points, parameters and demonstrations follow the source; the explanations and figures are newly written; the strip condition of two boundaries and the midpoint proof are [Added].